GREENFIELD DEVELOPERS PVT LTD
BIM Coordination — Request for Information

RFI #RFI-PRJ-2024-GT-001-CLH-001

Issued: 21 September 2026
Auto-generated by BIM Clash Engine

● CRITICAL SEVERITY | HARD CLASH | MEP vs STRUCTURAL | LEVEL 3 | GRID D4 / E4

PROJECT INFORMATION

Project Name Greenfield Tower — Block A	**Project Number** PRJ-2024-GT-001	**Client** Greenfield Developers Pvt Ltd
Location Sector 62, Noida, Uttar Pradesh	**RFI Number** RFI-PRJ-2024-GT-001-CLH-001	**Date Issued** 21 September 2026

RFI SUBJECT

RFI-001: Critical Hard Clash - HVAC Duct vs RCC Beam at Level 3 Grid D4/E4

CLASH ELEMENT DETAILS

Field	Element A — MEP	Element B — Structural
Revit ID	REV-MEP-003421	REV-STR-001187
Category	HVAC Duct	Structural Framing
Family	Rectangular Duct	RCC Beam 450x600
Size	800mm x 400mm	450mm x 600mm
System	Supply Air — AHU-03	—
Level	Level 3	Level 3
Grid Ref	D4 / E4	D4 / E4
Clash Point	X: 42.350 Y: 18.720 Z: 9.150	Penetration: -0.180m

TECHNICAL DESCRIPTION

A hard clash exists at Level 3, Grid D4/E4, where the 800mm x 400mm rectangular supply air duct from AHU-03 (Revit ID: REV-MEP-003421) penetrates the 450mm x 600mm RCC beam (Revit ID: REV-STR-001187) with a penetration depth of 180mm. This interference compromises the structural integrity of the beam and violates MEP routing clearances.

IMPACT ASSESSMENT

Failure to resolve this clash immediately will result in rework of structural concrete work, potential structural failure if unapproved penetrations are made, and significant schedule delays in the MEP installation phase for Level 3.

APPLICABLE CODES & STANDARDS

- IS 456:2000 (Plain and Reinforced Concrete - Code of Practice)
- NBC 2016 Part 4 (Structural)
- NBC 2016 Part 6 (Fire and Life Safety - Duct clearance)
- IS 800:2007 (General Construction in Steel - for reference on duct supports)

PROPOSED RESOLUTION

Reroute the HVAC duct to pass below the soffit of the RCC beam or through a pre-approved structural opening. If rerouting is not feasible due to spatial constraints, a structural penetration sleeve with appropriate reinforcement detailing must be designed and approved by the Structural Engineer.

Option 1: Reroute the duct to the adjacent grid line (D3 or E3) where vertical clearance is sufficient, maintaining a minimum 50mm clearance from the beam soffit.
Option 2: Install a structural penetration sleeve with additional shear reinforcement (stirrups) around the duct penetration, subject to structural engineer's calculation and approval.

INFORMATION REQUIRED

Please provide a revised MEP routing plan for the duct at Grid D4/E4 or approve a structural penetration sleeve with detailed reinforcement drawings. Confirm the minimum required clearance from the beam soffit for MEP elements.

ROUTING & RESPONSE DEADLINE

Assigned To	Role	Email	Response Required
Rajesh Kumar	Lead MEP Engineer	devendragupta56789@gmail.com	24 hours

Issued By: Devendra Gupta | BIM Automation Engineer
Company: Greenfield BIM Team | **Date:** 21 September 2026 | **Auto-generated:** 21-Sep-2026 09:34